Water storage change across Thailand

A basin assessment of climate and the remaining residual

Farzeen Khan

MS in Water Engineering and Management
Asian Institute of Technology

Advisor: Dr. Natthachet Tangdamrongsub

Why water storage matters

- Rainfall tells us how much water arrives. Storage tells us how much remains.
- Total water storage includes soil, rivers, reservoirs and groundwater.
- This study examines monthly changes across all 22 principal basins.

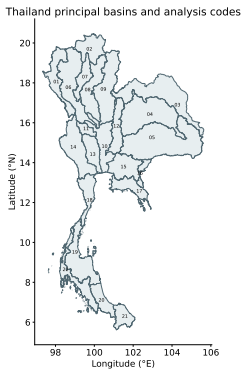
The problem is identifying the cause

- Storage can fall during drought, increased water use, or both.
- GRACE observes total water mass; it does not observe pumping separately.
- A model residual needs additional evidence before it can be called a human effect.

Three research objectives

Objective	Output
1 Storage change	Basin trends and monthly TWSA / GWSA
2 Climate and residual	Evaluated reconstruction and component checks
3 Decision support	Interactive basin website and runnable code

Thailand’s 22 principal basins



Study area · See the report for dates, sample counts and uncertainty.

Interpretation

Every basin is included in the maps and tables.

The polygons report basin averages; they do not create independent GRACE resolution.

What each dataset contributes

Data	Question it addresses
GRACE and storage components	How is total / groundwater storage changing?
Naturalized model and weather	Can climate-related storage be reconstructed?
GLDAS matching land storage	Do soil, canopy and snow estimates agree?
Wells and reservoir scenes	What independent component evidence exists?

Method 1 calculate storage and trends

- Groundwater = total − soil − surface − canopy − snow storage.
- Use common units, the 2004–2009 reference and valid spatial support.
- Seasonal Sen slopes describe the rate of change; Mann–Kendall tests assess ordered change.

A negative anomaly is storage below the reference, not a negative absolute water volume.

Method 2 test the climate reconstruction

Stage	Period	Purpose
Warm-up	1981–1990	Initialize weather memory
Training	1991–2009	Fit weather-storage relationships
Validation	2010–2015	Select models and weights
Testing	2016–2023	Evaluate unseen years

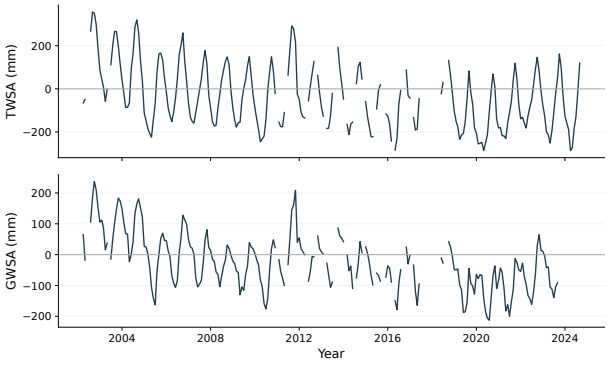
NSE: 1 perfect; 0 mean benchmark; negative worse. Anomaly NSE removes the training seasonal cycle.

Method 3 interpret the residual

- Residual = observed storage − naturalized climate baseline.
- It contains the model's human effect plus an unexplained observation-model gap.
- Check matching GLDAS compartments, groundwater levels and reservoirs separately.

The weather emulator reconstructs the naturalized parent. It does not independently observe the true climate-only counterfactual.

National storage declines



Results · See the report for dates, sample counts and uncertainty.

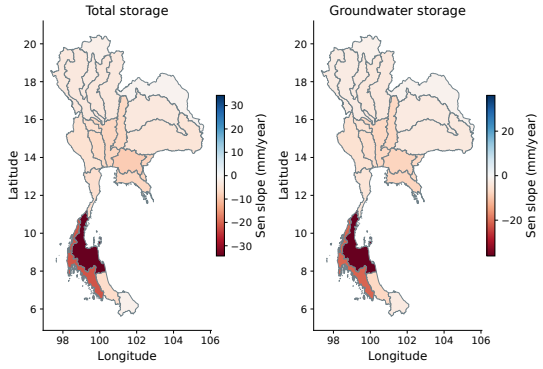
Interpretation

Total storage: −7.51 mm/year.

Derived groundwater: −6.84 mm/year.

Wet-year recovery can occur within a longer decline.

Trend magnitude varies across basins



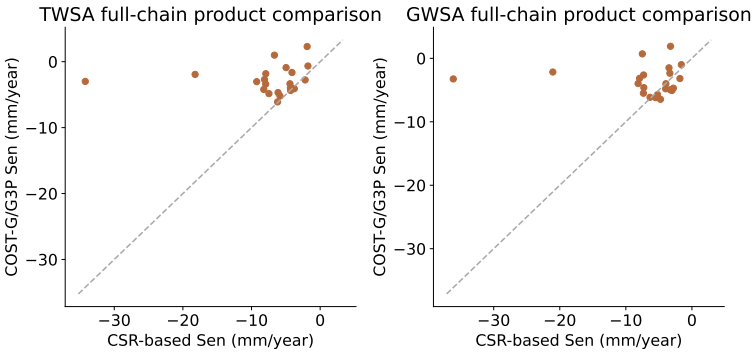
Interpretation

Negative colours indicate declining storage.

Use the map for regional screening; neighbouring basins share gravity support.

Results · See the report for dates, sample counts and uncertainty.

Basin rankings depend on the processing chain



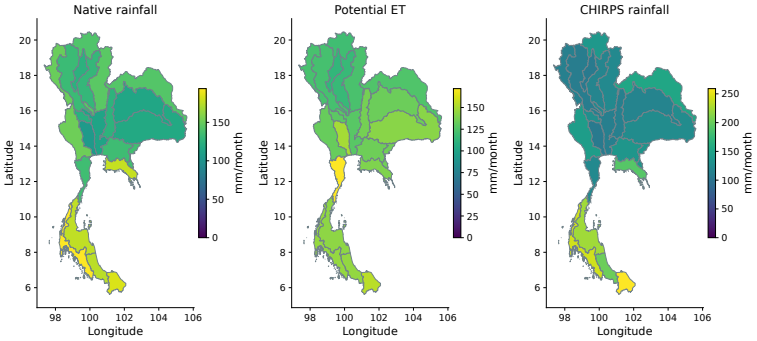
Interpretation

National decline persists in both chains.

Basin rank agreement is weak: 0.138 for total and 0.027 for groundwater.

Results · See the report for dates, sample counts and uncertainty.

Rainfall and atmospheric demand vary in space



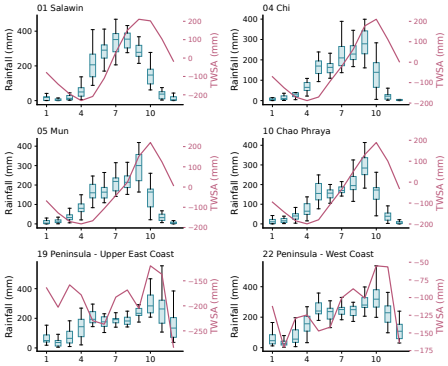
Interpretation

Rainfall and potential ET are the model inputs.

These are monthly fluxes; they are not stored water.

Results · See the report for dates, sample counts and uncertainty.

Storage responds to the seasonal water balance



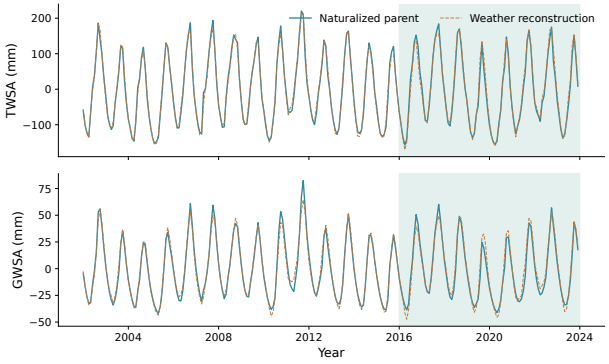
Interpretation

Blue boxes show rainfall by month. Pink shows mean storage on the right axis.

Storage integrates preceding rainfall and losses. Six geographical examples are shown.

Results · All 22 basins are retained in the report and numerical tables.

Weather reconstructs the naturalized parent



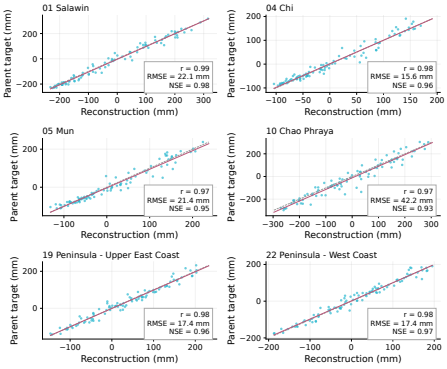
Results · See the report for dates, sample counts and uncertainty.

Interpretation

Shaded years are unseen test data.

High agreement evaluates the parent model; it does not prove climate-only observational accuracy.

Held-out predictions are close to their target



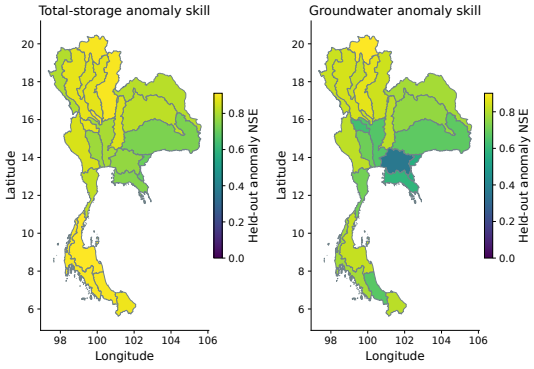
Results · All 22 basins are retained in the report and numerical tables.

Interpretation

The grey line is perfect agreement.

RMSE is error in mm; NSE compares error with a mean benchmark. All 22 basin scores are reported.

Anomaly skill is not spatially uniform



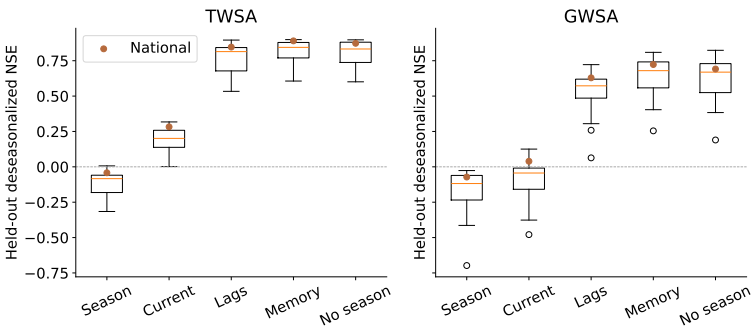
Results · See the report for dates, sample counts and uncertainty.

Interpretation

Higher NSE means better reconstruction.

Anomaly skill removes the usual seasonal cycle, giving a stricter test.

Climate memory improves reconstruction



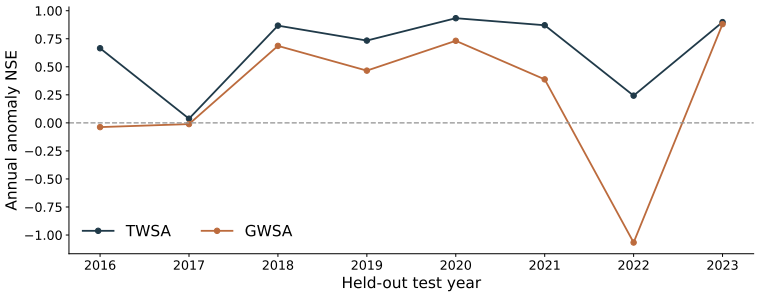
Interpretation

Past weather adds information beyond current rainfall and PET.

National anomaly NSE rises from 0.283 to 0.890 for total storage.

Results · See the report for dates, sample counts and uncertainty.

A pooled score can hide a weak year



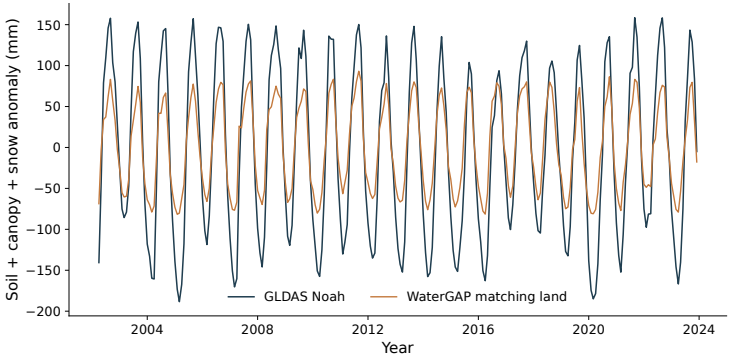
Interpretation

Groundwater anomaly NSE is −1.067 in 2022.

The model performs worse than the mean benchmark in that year; the failure is retained.

Results · See the report for dates, sample counts and uncertainty.

GLDAS checks matching land storage



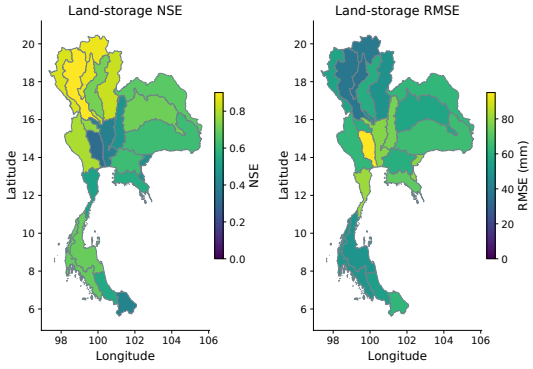
Interpretation

National r = 0.969; RMSE = 52.49 mm.

The comparison covers soil, canopy and snow only. It does not validate groundwater or human effects.

Results · See the report for dates, sample counts and uncertainty.

GLDAS agreement differs among basins



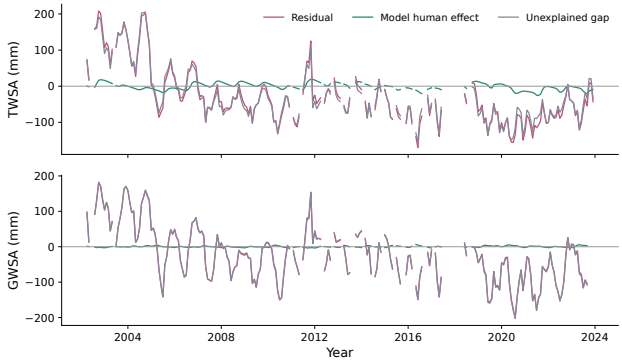
Interpretation

Higher NSE and lower RMSE mean closer agreement for the same land-storage target.

Shared climate and component dependence limit independence.

Results · See the report for dates, sample counts and uncertainty.

The residual is larger than the modeled human effect



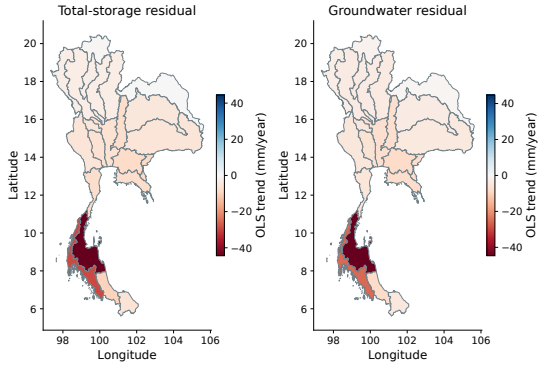
Results · See the report for dates, sample counts and uncertainty.

Interpretation

The residual closely follows the unexplained gap.

The entire residual cannot be treated as measured human depletion.

Residual trends identify where to investigate



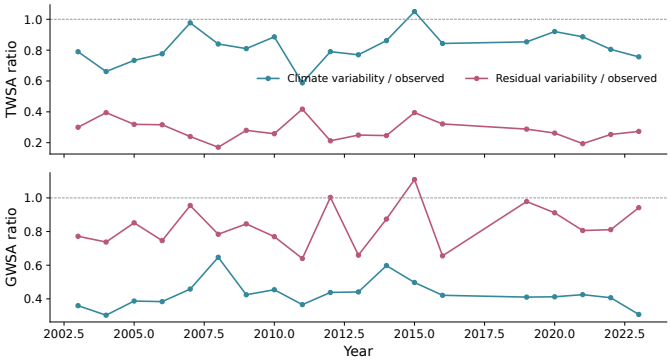
Results · See the report for dates, sample counts and uncertainty.

Interpretation

Negative rates mean observations decline relative to the naturalized baseline.

Human influence and model or measurement errors can both contribute.

Variability changes through time



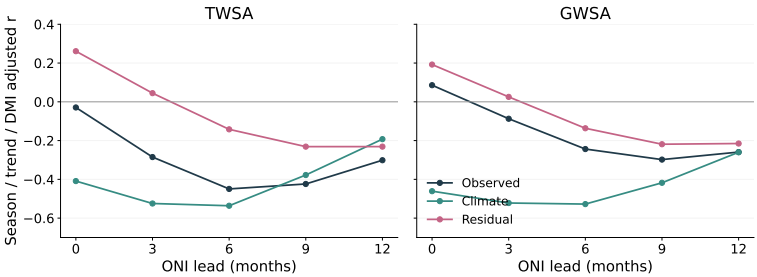
Interpretation

Climate and residual variability differ from year to year.

These ratios need not add to one. They are not human-versus-climate percentages.

Results · See the report for dates, sample counts and uncertainty.

ENSO is clearer in the climate baseline



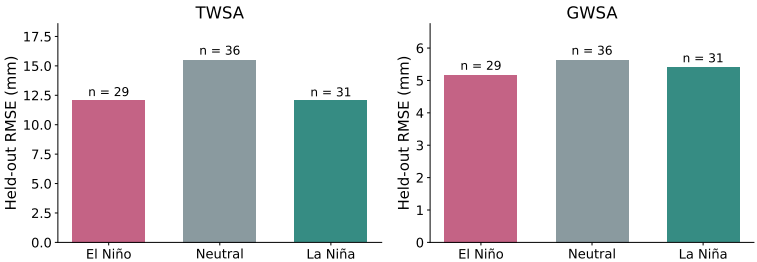
Interpretation

At six months, baseline r is about −0.53; residual r is about −0.14.

No basin test passes the conservative FDR screen. This does not prove no remaining climate effect.

Results · See the report for dates, sample counts and uncertainty.

ENSO phase scores remain conditional



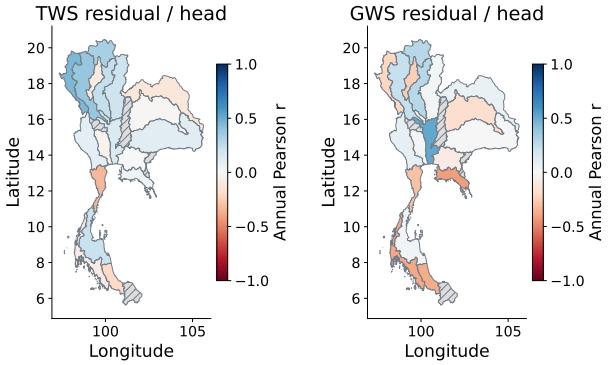
Interpretation

The model is frozen across all ENSO phases.

Months share episodes, so these are reconstruction comparisons rather than independent event forecasts.

Results · See the report for dates, sample counts and uncertainty.

Groundwater support is weak and uneven



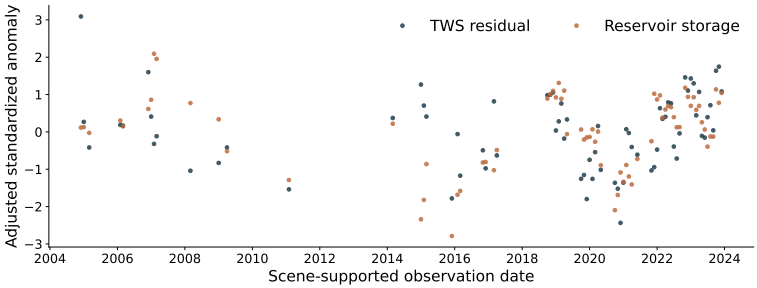
Results · See the report for dates, sample counts and uncertainty.

Interpretation

National groundwater-residual / head r is only 0.037.

Head is a local water level, not regional water volume. The balanced-panel result is near zero.

Reservoirs support a surface-water component



Interpretation

Adjusted r = 0.439 across 73 scene-supported months.

Reservoirs respond to climate and operations together; this cannot validate the total human residual.

Results · See the report for dates, sample counts and uncertainty.

Conclusion what the study establishes

- The primary national records show declining total and groundwater storage.
- Weather memory reconstructs the naturalized model well, with identifiable weak years.
- The residual remains human influence plus unresolved discrepancy; independent evidence is limited.

Discussion how to use these results

- Storage detection is stronger than precise human attribution.
- Product choice and monitoring coverage change local interpretation.
- Use basin results to prioritize investigation and monitoring, supported by the downloadable evidence.

Further work that would strengthen the study

- Obtain consistent monthly wells, aquifer properties and pumping records.
- Improve surface-water observations and matched GRACE spatial response.
- Test alternative naturalized models and longer climate-event records.

References and reproducible research

- Ji et al. (2025), climate reconstruction: doi:10.1080/17538947.2025.2557516
- Müller Schmied et al. (2024), WaterGAP: doi:10.5194/gmd-17-8817-2024
- Chueasa et al. (2024), Thai rainfall and ENSO: doi:10.1016/j.mex.2024.102954
- Sharifi et al. (2025), spatial response: doi:10.5194/hess-29-6985-2025
- Mahto et al. (2025), reservoirs: doi:10.5194/essd-17-2693-2025

Code and full bibliography: github.com/pratikkhanal16/thailand-water-storage-dss. Metropolis theme: matze/mtheme.